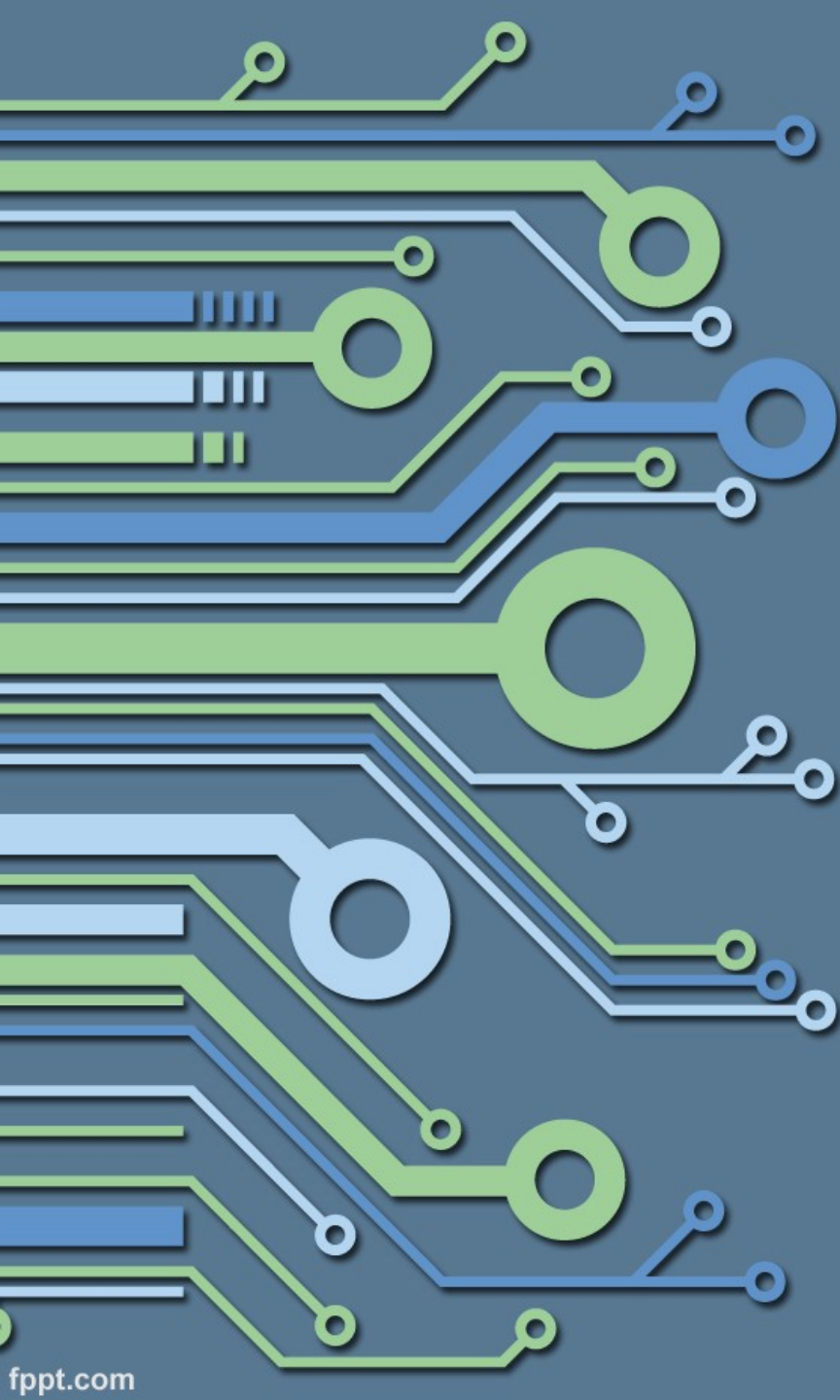




Wardriving with Microcontrollers

CoD_Segfault



Who Am I?

- CoD_Segfault
- WiGLE rank: 39 (as of 2025-07-11)
- Author of BW16-Open-AT
- Code contributor to wardriver.uk
- Hardware designer
- Hard Hat Brigade Organizer
 - Analog Synth Hat
 - DOOM Hat

What is Wardriving?

Wardriver

@SecureThisNow

\ wôr'drīvər \ *noun*

person who spends an
unreasonable amount of
money and time on the hobby
of searching out and mapping
wireless networks

Also see: stumbler, wiggler

Image credit: MrBill

Traditional Wardriving

- PC with 1 or more Wi-Fi adapters
- GNSS (GPS, BeiDou, GLONASS, Galileo)
- Software (Kismet, inSSIDer, etc.)

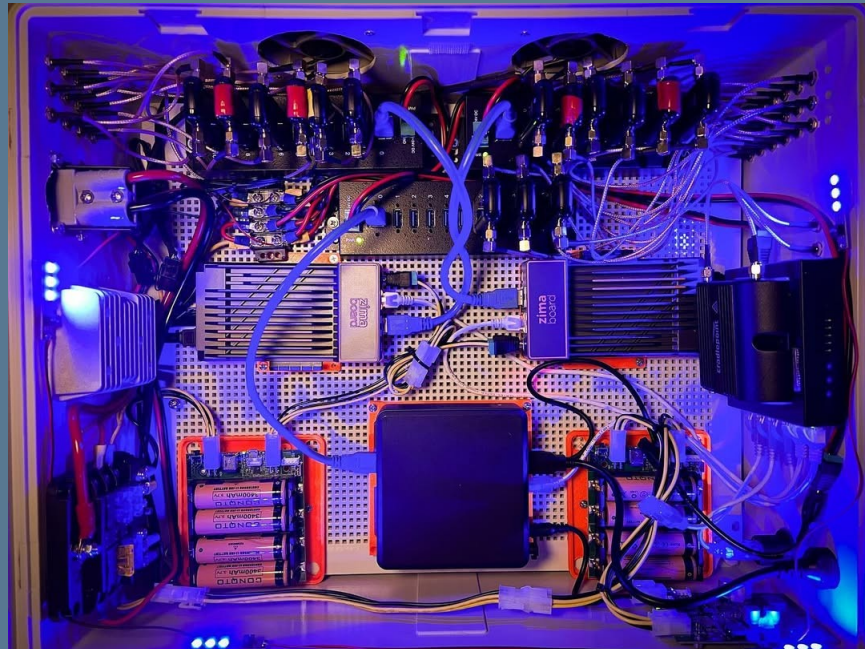
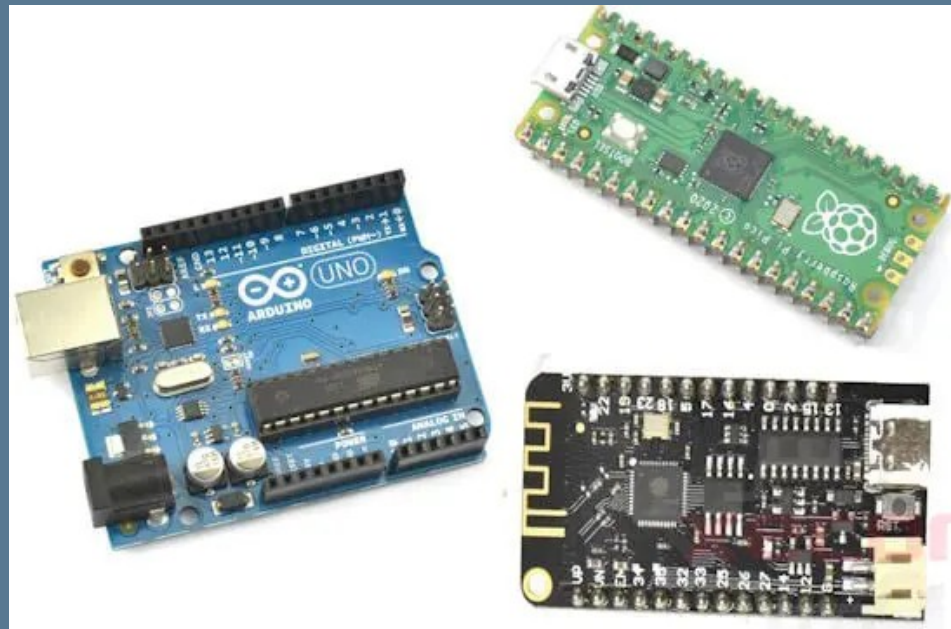


Image credit: BusySignal

What is a Microcontroller?

- Small computer on a single chip.
- Often has many peripherals built in (SoC).
- Many now contain Wi-Fi



A decorative graphic on the left side of the slide, consisting of a vertical arrangement of stylized circuit traces. These traces are colored in shades of green and blue, with some circular nodes and branching lines, resembling a printed circuit board (PCB) layout.

Why use a Microcontroller?

- Low Power requirements.
- No OS overhead.
- Boots in seconds.
- Easy to run multiple.
- Inexpensive.

How it Started (wardriver.uk)

- Joseph Hewitt made the first version in 2019.
 - Arduino with an ESP8266 running AT firmware.
- Version 2 was 2x ESP8266 chips.
- Version 3 upgraded to 2x ESP32 chips.
 - Added Bluetooth, GSM.
 - Later replaced GSM with BW16.

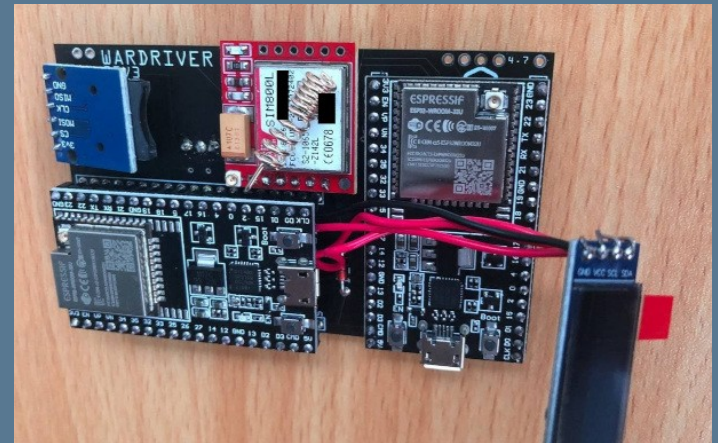
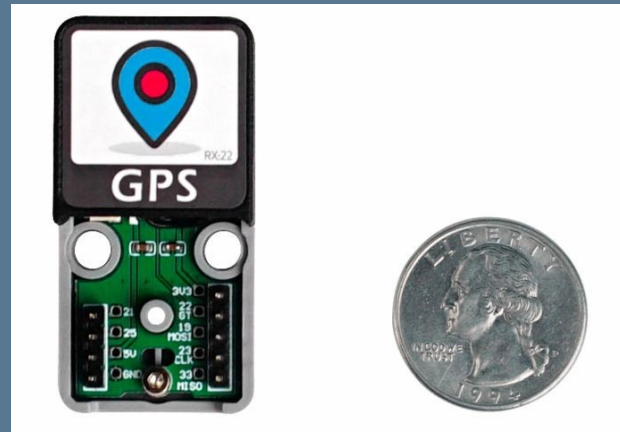


Image credit: Joseph Hewitt fppt.com

M5 Atom GPS

- Originally implemented by Lozaning
- Major code contributions by Luke Switzer
- Ultra compact



WiFydra

- Designed by Lozaning
- Up to 16x ESP32-C3 Or ESP32-S3 set to each scan individual channels.

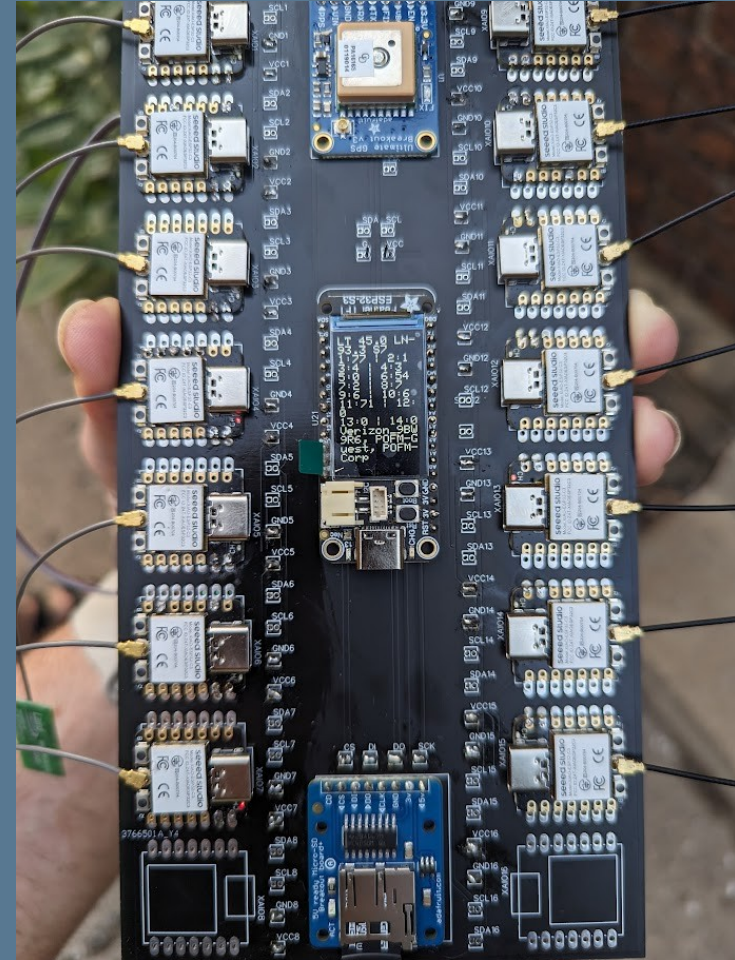
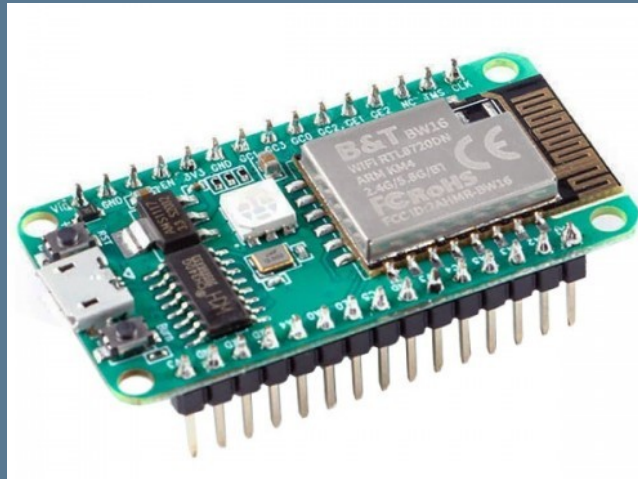


Image credit: Lozaning

BW16 – 5GHz Support

- Manufactured by B&T
- Realtek RTL8720DN
- Shipped with an AT firmware
- Newer modules have different AT commands



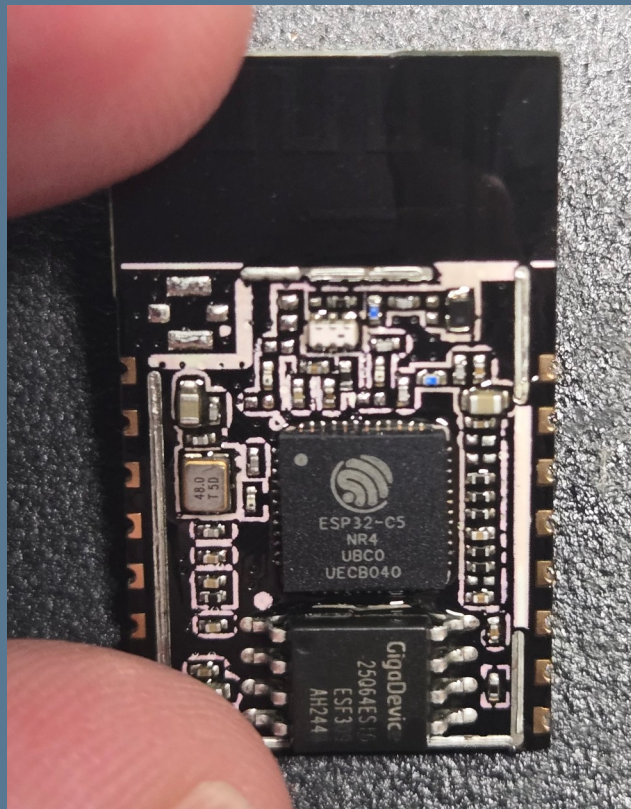


BW16-Open-AT

- Written to be an open replacement for the proprietary B&T firmware.
- Only implements functions needed for wardriver.uk wardrivers.
- Better handling of encryption

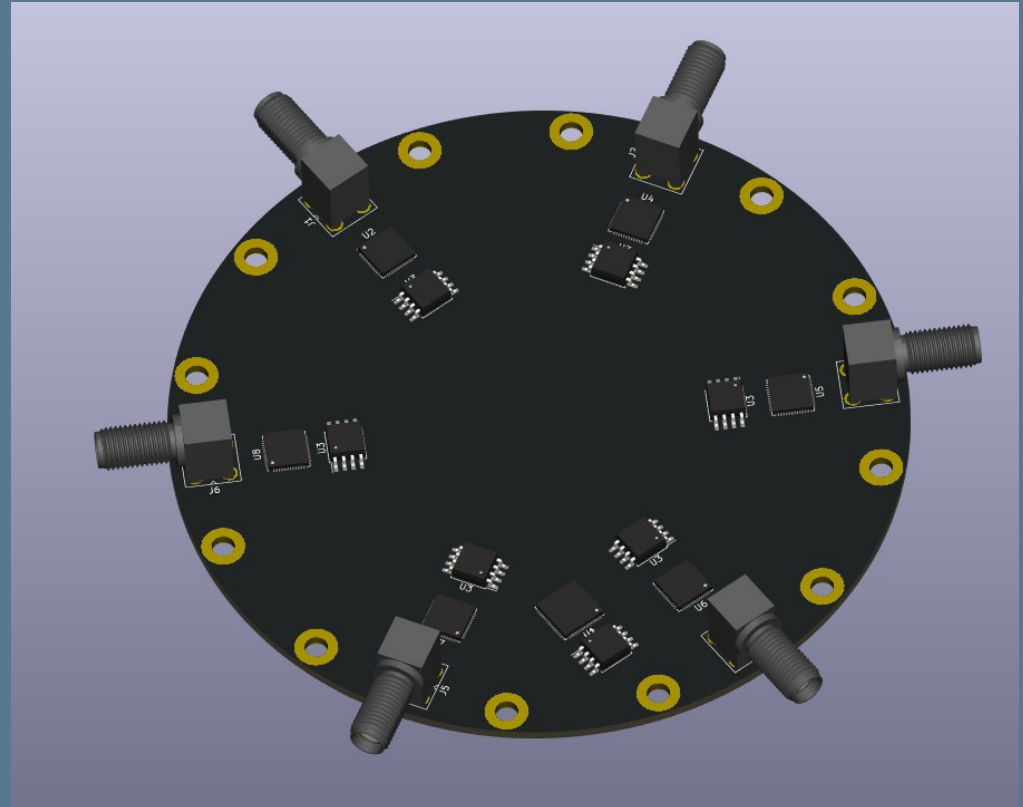
The Future (ESP32-C5)

- First dual band espressif microcontroller.
- (Mostly) code compatible with earlier versions.



WiFi Shuriken (In Dev)

- 6x ESP32-C5
- 1x RP2350
- GPS
- MicroSD





Questions? Contact

- Bluesky: @se.gfau.it
- Mastodon:
@CoD_Segfault@infosec.exchange
- Email: contact@se.gfau.it